Phase B — the beam in the loss

Status: **DEFERRED / SUPERSEDED FOR EVENTS BY `eb22g` (`PAPER_3_DOCUMENT_EVENTS_AND_COREF.md`); OPEN ONLY FOR RELATIONS & EXCLUSIVE RECORD FIELDS.** *(Updated 2026-11-06. Written 2026-09-15 after Phase A returned; between September and November 2026 the event line solved its binding, clustering, and proposal bottlenecks directly inside the boundary record head via `event_records: true`, the relational junction layer `record_link_mode: junction`, `TriggerLink` / `EntityCorefLink` coreference, and balanced positive-negative supervision in `eb22g` — lifting full-mix validation `event_argument` strict micro-F1 from **`0.0627`** to **`0.3046`** [relaxed **`0.5342`**, fair **`0.5221`**], `event_trigger` to **`0.5720`**, and 20,602-record blind-test per-head `event_argument` strict micro-F1 to **`0.4637`** [relaxed **`0.5985`**, fair **`0.5901`**] and `event_trigger` to **`0.6994`** [`0.7227` `@0.9`] without a training-time beam. See §10 and [[EXPERIMENT_CATALOG]].)* Companion to [[JOINT_IE_DESIGN_RECORD]] §7 (which this expands), [[RESEARCH_PROGRAM]] §3, and [[PAPER_3_DOCUMENT_EVENTS_AND_COREF]].

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1. Why this is now the interesting arm, and not a consolation prize

The programme's claim is that dense document-level extraction needs **global structured inference over the boundary candidate scores**, not the greedy per-chunk decode. Phase A tested the decode-time half of that and came back **null at parity**: all seven heads inside the ±0.02 floor, at ~2.5× the wall clock.

That is a real negative and it is reported as one. But it is a null about a beam that has **never been trained for**, and — as of 2026-09-08 — one that had **never been decoded with its own constraint machinery live** either. Cardinality selects the joint beam's utility (`logit − absent` against bare `logit`) and its exclusivity slot, and every `dtype: str` field compiled `ZERO_OR_MORE` because a parameter no caller passed. The scalar half of decision B was inert for the entire history of the measurement.

So the honest statement of where the thesis stands:

version of the claim	status
decoding **with** the beam beats greedy decode	**tested, null at parity** (Phase A, corrected)
decoding with the beam, constraints actually live	**tested once**, same null
training **for** the beam beats training for greedy	**never tested** — this document

Phase B is the only untouched version. It is also the version the literature would recognise: Phase A compared two decoders over one model trained for neither.

A null here would close the programme's central question properly. That is worth saying plainly: this plan is not written expecting a win. It is written because the question is currently unanswered, and because a negative from a trained-for beam is a much stronger result than a negative from a decode swap.

2. What Phase A settled, and what it did not

Settled. Over one trained model, swapping `decode_mode` moves nothing outside the noise floor on any head, on two independent lineages (`eb16-rebuild-tr` and `joint-boundary-mmbert-137k-clean`), and beam width is not a lever — a 4/16/64 sweep moves structure by 0.0018 and *narrower* is marginally better.

Not settled. Whether a model whose loss is computed through the beam learns different candidate scores. Phase A holds the model fixed by construction; it cannot answer this.

A trap this document exists partly to prevent: the width sweep's flatness is sometimes read as "search capacity does not help, therefore joint inference does not help". It does not support that. W=1 barely searches and wins, which says the working contrast is **independent thresholding vs constrained joint selection**, not *greedy vs beam*. Phase B tests the formulation, so it should be built and reported in those terms.

3. The mechanism — and why it is a training-loop change, not a new component

The boundary record head already uses the idiom Phase B needs:

> **assignment on DETACHED scores, loss RECOMPUTED differentiably on the matched pairs.**

In `records.py::compute_dense_batch_loss` the Hungarian matching runs on `cost.detach().cpu()` — a hard, non-differentiable assignment — and the resulting index pairs are then used to recompute a differentiable loss from the live logits. Gradient flows through the *scores*, never through the *argmin*.

Phase B replaces **which assignment** that step produces:

```
                  today (Phase A + greedy training)     Phase B
assignment        Hungarian over a per-group cost       joint_ie beam over the whole
                  matrix, detached                      JointProblem, detached
loss              recomputed on matched pairs           recomputed on beam-selected
                                                        pairs, identically
gradient path     through candidate/assign logits       unchanged
```

Two consequences worth stating because they cut the work down:

1. **No new autograd machinery.** The beam is already a pure scoring-and-selection pass over `candidate_scores`; it needs no gradient. The only requirement is that it return index pairs in the same shape the matcher returns today.
2. **One objective covers BOTH faces.** An event's role edges and a relation's plain edge are the same `EdgeCandidate` in the same `JointProblem` (`RESEARCH_PROGRAM` §1), so events and relations are trained by one mechanism rather than two.

The open design question, and it is not small: a structured-prediction objective needs a *contrast*, not just a positive. The cheapest defensible form is a **structured hinge / margin**: the loss pushes the gold structure's score above the beam's best wrong structure by a margin, which is exactly what the beam already computes.

The alternative — treating the beam's selection as a soft target — needs a temperature and a second hyperparameter and should not be the first thing tried.

4. Decide FILL vs REJECT before writing any code

Carried from `JOINT_IE_DESIGN_RECORD` §7 and repeated here because it is the one decision that silently invalidates the comparison:

`decode_group`'s scalar path runs `if chosen is None or chosen == 0: continue` **before** the cardinality check, so when `ABSENT` wins the argmax a `REQUIRED_ONE` field is simply left empty — exactly as an optional one is. Greedy **fills**; it never rejects an instance for an unfilled required role.

Building `RequiredRoles` as a **rejection** constraint would therefore make the beam stricter than the decoder it is measured against, and the resulting precision/recall gap would read as a property of global decoding rather than of the constraint. **Only `fill` preserves comparability.** If a rejection variant is ever wanted, it is a third arm, not a default.

5. Pre-registered bars

Written before the run, per the standing rule, and stated so a null is publishable.

Primary. `relation` and `event_argument` strict micro F1 on the shared blind test, against a control trained identically with the greedy objective. The bar is **> ±0.02** (the measured single-run floor). Relation's own floor is **±0.041** on `mix_natural`, so a relation claim needs either that margin or a second seed — and the honest choice is the second seed.

Secondary, and read as a cost not a win. Wall-clock per epoch and the record head's throughput (§7). A treatment that wins by 0.03 and trains 5× slower is a result, not a recommendation.

Guard heads. `entity`, `event_type`, `classification`. The structured objective touches edges; if the entity head moves materially, the change has leaked and the run is uninterpretable rather than positive.

Two seeds per arm, minimum. Every one-seed curve in this project is suspect (`TODO.md`), a control re-run of a published recipe once scored +0.023 above it, and this programme has already retracted findings for exactly this reason.

6. The instrument requirements, which are not optional here

Three boxes and ~$4.20 were spent in September 2026 on runs that measured nothing, and not one failure was the mechanism under test. Phase B is a bigger spend and inherits the rules those failures bought:

- **The run must PROVE the treatment executed**, in its own log, before any metric is read. For Phase B that means printing which assignment path produced the matched pairs, per arm, per epoch — not inferring it from the numbers the change was supposed to move.

- **The gate must be able to FAIL.** A gate that samples one batch is not a gate; it must be deterministic and per-arm. Both arms print it, because a line that appears only in the treatment proves nothing about the control.
- **Verify on the TRAINING path, never on inference.** Three separate defects in September were "verified on the inference path, assumed training matched". Phase B is a training change; every check belongs in `collate_fn_train` and the loss, not in `extract`.
- **`GLINER2_STRICT_ATTN=1`, and confirm it can still raise.** On bf16 ModernBERT an sdpa fallback is a correctness failure, and the guard was unreachable by construction until 2026-09-15.

7. The risk most likely to kill this, and it is not the science

The record head trains at ~4.6 samples/s against the curve's 22 on the same H100 and the same model (`JOINT_IE_DESIGN_RECORD` §7, 2026-08-10, unresolved). The signature — idle GPU, one pegged core, idle dataloader workers — is many small GPU kernels behind heavy Python-side work, and the record path is the prime suspect by elimination.

Phase B *adds* a Python-side selection pass per batch. If the beam is run naively inside the training loop, the sensible prior is that it lands somewhere between 5× and 10× the greedy objective's wall clock, which prices a real comparison out of reach.

So the first increment is a throughput probe, not an experiment: run the beam over one batch's `JointProblem` at training shapes and time it against the Hungarian matcher it replaces. If it is worse than ~2× the matcher, the plan changes — batch the beam across groups, or cap it — before any arm is trained. **Do not skip this to "just try it".**

8. Cost model

step	what	cost
B0	Throughput probe, one batch, CPU or one GPU-minute	**~$0**
B1	Wire beam-assignment into the loss behind a config flag; unit-test that it returns matcher-shaped pairs	$0
B2	Smoke: 200 steps, both arms, one seed — does loss descend, does throughput hold	~$2
B3	Real arms: 2 seeds × 2 arms on a mid-size corpus	~$25–40
B4	If positive: repeat on the second lineage before quoting	~$25

B0 through B2 are ~$2 and gate everything after. Nothing beyond B2 should be booked until B0's number is known.

9. What would falsify the thesis

Stated so the negative is as publishable as the positive:

> Trained-for-the-beam matches greedy-trained-and-greedily-decoded on relation and event_argument, within the noise floor, at two seeds — while costing materially more wall clock.

That result, combined with Phase A's decode-time null, would say the programme's claim is **not supported at the scales and corpora tested**, and would localise the remaining possibility to regimes this project has not reached (denser documents, larger candidate sets). That is a legitimate contribution and should be written up as one rather than buried.

10. Open questions this plan does not answer

- **Which corpus.** Re-DocRED is the dense-relation case the thesis is about; RAMS is the event case. Doing both doubles B3. Undecided.
- **Margin form.** Structured hinge is proposed above as the cheapest defensible contrast, not as a settled choice.
- ~~**Whether the cardinality regime change should land first.**~~ **ANSWERED 2026-09-15: it does not need to.** The A/B came back NULL -- structure strict 0.7388 -> 0.7326, -0.0061, inside the floor, with precision up 0.020 and recall down 0.022 cancelling. So the record head's loss is NOT a moving target underneath Phase B, and the sequencing question dissolves. One seed on one structures-only corpus, so this closes the *sequencing* worry rather than the regime question itself.
- **Whether `event_records` changes what Phase B is measuring — ANSWERED 2026-11-06 (`PAPER_3_DOCUMENT_EVENTS_AND_COREF.md`).** When this section was written on 2026-09-15, `eb16-eventrecords-tr` was just launching and §3 above assumed record assignment ran through `compute_dense_batch_loss` (Hungarian matching). In the `eb17` → `eb22g` architecture that actually shipped: 1. **Events use `natural` (`per_query`) mode (`compute_group_loss`), not Hungarian matching:** Each trigger candidate seeds an instance directly (`record_coreferent_ownership: true`), arguments are scored by the gated bilinear + geometry **junction layer** (`record_link_mode: junction`) and supervised by a contrastive per-column trigger softmax (`_column_loss`) plus split positive/hard-negative BCE (`_list_field_bce`), and coreferent triggers are merged by `TriggerLink` (`record_coref_link: true`). 2. **The event binding and proposal bottlenecks were solved without a beam in the loss:** Fixing `_list_field_bce` `(1+K)` dilution (`eb22f`) and `candidate_pair_loss` absent-query 7,700× dilution (`eb22g`: `pair_split_reduction: true`, `hard_negative_keep_all_when_absent: false`, occurrence/coref target grouping) lifted 64-doc overfit `@0.5` argument strict micro-F1 to `0.4319` on CASIE and `0.8005` on CMNEE, and lifted full-mix Epoch 4 validation `event_argument` strict micro-F1 to **`0.3046`** (relaxed **`0.5223`**, fair **`0.5080`**) and `event_trigger` to **`0.5656`**. 3. **The record-head throughput bottleneck in §7 was also fixed:** Removing per-field `torch.nonzero` GPU-CPU syncs in `forward_group` (`8d9f6f8`) and disabling cuDNN SDPA recompilation restored single-A100 training

throughput to `8.2–9.0 samples/s`. 4. **What remains for Phase B (if ever pursued on top of `eb22g`):**

- **Document-level relations (`Re-DocRED`):** Where `W=1` constrained joint selection already beat greedy by `+0.052` strict micro-F1 at decode time (`JOINT_IE_SCALING.md` §4.1), a structured hinge loss over `W=1` joint relation selection remains a clean, self-contained experiment.
- **Cross-instance exclusivity (`exclusive=True`) on multi-instance structures:** Where Hungarian assignment Across instances prevents two `casualty_report` records from claiming the same number (`run_pipeline.py:390`).